Seminar 06 cheat sheet

Second-Order Methods, Part Two

Optimization Methods
HSE University, Faculty of Computer Science, 2026–2027
Course lecturer: Andrey Ignatov
Author: Evgeny Baulin, Telegram @tarakan_tuc

Gauss–Newton

- $f = \frac{1}{2}\|r\|^2$: $\nabla f = J^\top r$, $\nabla^2 f = J^\top J + \sum r_i \nabla^2 r_i$.
- GN: $J^\top J p = -J^\top r$. Needs full column rank of J.
- Small residuals near x^*: quadratic convergence; big residuals: linear.

Levenberg–Marquardt

- $(J^\top J + \lambda I)p = -J^\top r$; $\rho = \frac{f - f^+}{m(0) - m(p)}$.
- $\rho \geq 3/4$: accept, $\lambda \mathrel{/}= 10$. $\rho < 1/4$: reject, $\lambda \mathrel{*}= 10$.
- Small λ: Gauss–Newton. Large λ: short gradient step.

Trust region

- $\min m(p)$ over $\|p\| \leq \Delta$: $(B + \lambda I)p = -g$ with $\lambda \geq 0$.
- Dogleg: broken line $0 \to p_C \to p_N$; farthest point in the ball.
- $\rho > 3/4$ on boundary: $\Delta \leftarrow 2\Delta$. $\rho < 1/4$: $\Delta \leftarrow \Delta/4$.

L-BFGS

- Store m pairs (s_i, y_i); $\gamma_k = s_{k-1}^\top y_{k-1} / (y_{k-1}^\top y_{k-1})$.
- Two-loop: back-sweep a_i, q; scale $\gamma_k q$; forward sweep b_i, r.
- One direction: $4mn$ work, $2mn$ memory, no $n \times n$ matrix.

© 2026 Evgeny Baulin, t.me/tarakan_tuc MIT License